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G. Farmer John's Last Wish

time limit per test: 3 seconds
memory limit per test: 256 megabytes

Bessie has found an array a of length n on the floor. There appears to be a handwritten note lying next to the array, seemingly written by Farmer John. The note reads:

Help me, dear Bessie! Let $f(a)$ denote the maximum integer k in the range $[1, n]$ such that $\gcd(a_1, a_2, \dots, a_k) > \gcd(a_1, a_2, \dots, a_{k+1})$, or 0 if no such k exists.

Bessie decides to help FJ. She defines $g(a)$ to represent the maximum value of $f(a)$ over all possible reorderings of a .

Bessie decides to not only find $g(a)$, but also the value of $g(p)$ for all prefixes p of a . Output n integers, the i 'th of which is $g([a_1, a_2, \dots, a_i])$.

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains an integer n ($1 \leq n \leq 2 \cdot 10^5$).

The following line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output n integers on a new line: the i 'th of which should be $g([a_1, a_2, \dots, a_i])$.

Example

input	Copy
3 8 2 4 3 6 5 7 8 6 6 6 6 6 6 6 9 8 4 2 6 3 9 5 7 8	
output	Copy
0 1 2 3 3 3 4 5 0 0 0 0 0 0 0 1 2 2 4 4 4 4 5	

Codeforces Round 1050 (Div. 4)

Contest is running

01:04:53

Contestant



→ Submit?

Language: Python 3.13.2
Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

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